



Revealing the activity of different iron carbides for Fischer-Tropsch synthesis

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ABSTRACT

Exploring the structure of iron-based catalysts on the catalytic performance of Fischer-Tropsch synthesis (FTS) reaction has attracted much attention. With this in mind, the mixture of SiO_2 or Al_2O_3 powder and non-porous iron oxide powder ($\alpha\text{-Fe}_2\text{O}_3$) have been investigated for understanding the nature role of different iron carbides in the FTS reaction by adjusting the formation of iron carbides. Under the typical FTS reaction conditions, the CO conversion of $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst could reach up to 61.6 %, which is about 3.3 times that of the pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst. Based on the characterization results, including in situ XPS and CO-DRIFTS as well as Mössbauer spectra, it is found that the electronic state of iron atoms is affected by the existence of SiO_2 or Al_2O_3 , and the interactions of Fe-Si or Fe-Al are formed on the surface of iron powder, which plays an important role in formation of C-rich iron carbide active phase ($\epsilon\text{-Fe}_2\text{C}$). Although $\chi\text{-Fe}_5\text{C}_2$ is usually as the active phase in the FTS reaction, we have found that the content of $\epsilon\text{-Fe}_2\text{C}$ is more positive to activity.

1. Introduction

Fischer-Tropsch synthesis (FTS) is regarded as a promising process of direct converting syngas (CO/H_2) from biomass, natural gas or coal to valuable chemicals and fuels [1–7]. Iron-based catalysts are generally recognized as potential catalysts compared to other catalysts due to the advantages of high activity, low cost and high selectivity [5,2–7]. It is generally accepted that the iron carbide phases are the active phase during the FTS reaction. Over the past years, researchers were focused on improving the performance of iron-based catalysts and studying the correlation between the iron carbide phases and the catalytic performance during the FTS reaction. Although tremendous effort has been made, the role of iron carbide phases during the FTS reaction, including their formation, and effects on reaction performance, is still controversial until now [8–10].

Up to now, researchers found and discussed many kinds of iron carbides, those can influence the reaction performance of FTS. Typically, $\chi\text{-Fe}_5\text{C}_2$ is considered to be the predominant active phase during the FTS reaction [11–14]. Recently, the pure $\chi\text{-Fe}_5\text{C}_2$ was been synthesized and proven to be positive for activity during the FTS reaction [14]. As reported by our previous studies, the C-poor iron carbides phase $\theta\text{-Fe}_3\text{C}$ was advantageous to forming light olefins from syngas via FTS reaction [8,15]. Additionally, the Fe_7C_3 was observed in CO-treated Fe/SiO_2 catalyst, and the investigation has elucidated that

the highest intrinsic activity was obtained during the FTS [16]. Interestingly, studies also indicated that the Fe_7C_3 was detected only over the SiO_2 -supported iron catalysts [16,17], implying the properties of support play an irreplaceable role in the formation of iron carbides phase. With the development of FTS, the C-rich $\epsilon\text{-Fe}_2\text{C}$ phase was observed by some researchers. It was found [18] that $\epsilon\text{-Fe}_2\text{C}$ phase closely relates to the stability and low CO_2 selectivity of the FTS catalysts. Park et al. [19] reported a carbon encapsulated iron-carbide nanoparticle supported on nitrogen-doped porous carbon ($\text{Fe}_5\text{C}_2@\text{C}/\text{NPC}$ or $\text{Fe}_3\text{C}@\text{C}/\text{NPC}$) catalyst, in which the $\epsilon\text{-Fe}_2\text{C}$ phase was detected after 138 h FTS reaction. Meanwhile, Wang et al. [20] found that the $\text{FeMnCu}/\text{h-Boron-Nitride}$ catalyst only existed the $\epsilon\text{-Fe}_2\text{C}$ phase after FTS reaction, and they cannot compare the effects of different iron carbides. Therefore, the understanding of C-rich $\epsilon\text{-Fe}_2\text{C}$ phase function is still confused during the FTS reaction due to the complex system, such as special preparation method, supports and multiple promoters. Moreover, it is well known that the abundant iron carbide phases ($\chi\text{-Fe}_5\text{C}_2$, $\theta\text{-Fe}_3\text{C}$, Fe_7C_3 , $\epsilon\text{-Fe}_2\text{C}$) often coexist more than one phase under reaction conditions over a conventional catalytic system [8,15], which make it difficult to identify exact role of each iron carbide phase during the FTS.

On the other hand, the investigation has also elucidated that the properties of the support are critical to the catalysis performance of the FTS reaction [21–24]. In general, the oxide supports have a significant

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effect on the dispersion or morphology of active metal, which would influence the electron properties of the surface active atoms leading to different FTS performance [25]. Meanwhile, the appropriate pore size is beneficial to the diffusion of reactants and products, and different pore size of catalysts would contribute to totally different reaction performance of the catalyst [6]. In addition, many studies found that the solid solution compound (iron silicate, iron aluminate) could be formed on the conventional oxide supports (SiO_2 , Al_2O_3), which may lead to worse reducibility of catalysts and carburization of the supported iron, resulting in lower FTS activity [13,22]. In contrast, the weak interaction between the iron species and support is advantageous to form of more active sites for the further carburization [26]. According to the above discussions, it is considered that the performance of FTS catalysts is affected by several points, such as structure of active phases, properties of supports and diffusion limitation, one of which may have a significant effect on the performance of the iron-based catalyst, making it difficult to determine what the nature role of iron carbides play in the catalytic performance.

Despite tremendous effort being made, the activity of different iron carbides is still unclear. Hence, it is essential to design an appropriate catalytic system to further understand the role of the iron carbides on activity of the FTS reaction. In this work, to investigate the FTS activity of different iron carbides, the nonporous iron-based catalysts, a physical mixture of iron powder ($\alpha\text{-Fe}_2\text{O}_3$) and powder of Al_2O_3 or SiO_2 , are applied to FTS reaction, aiming to avoid the influence of morphology of active phase, porous structure and multiple promoters. Combining the characterization results, including in-situ XPS, in-situ CO-DRIFTS and Mössbauer spectroscopy, we have found that the electronic state of surface iron atom and the phase of iron carbides were affected by the presence of Al_2O_3 and SiO_2 , and the content of $\epsilon\text{-Fe}_2\text{C}$ is more positive to activity of FTS.

2. Experimental section

2.1. Materials

Iron powder ($\alpha\text{-Fe}_2\text{O}_3$) is purchased from Aldrich-sigma. Silica (amorphous) and Aluminum oxide ($\alpha\text{-Al}_2\text{O}_3$) are supplied by Beijing Chemical Corp.

2.2. Catalyst preparation

The $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst (mass ratio, the same below) and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst were prepared by physical mixing approach. The $\alpha\text{-Fe}_2\text{O}_3$, SiO_2 , Al_2O_3 were dried for 12 h at 120°C to remove water. After that, three catalysts of $\alpha\text{-Fe}_2\text{O}_3$, $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ were ground in a mortar for 30 min, respectively. Then, all samples were pressed to 8 MPa, crushed and selected 20–40 mesh particles. The selected three catalysts of $\alpha\text{-Fe}_2\text{O}_3$ (0.5 g), $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ (1.0 g) and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ (1.0 g) together with 1 g, 0.5 g, 0.5 g quartz sand (20–40 mesh particles), respectively, were used for Fischer-Tropsch synthesis (FTS) reaction. In this way, we can guarantee that the content of iron oxide and the height of the catalyst bed are consistent for the three catalysts during the FTS reaction.

2.3. Catalyst characterization

In order to obtain the reduced catalysts for characterization, the fresh catalysts were reduced with syngas ($\text{H}_2/\text{CO} = 1$, 30 mL min^{-1}) for 10 h at 350°C , then switch to 1% $\text{O}_2/99\%$ Ar at the room temperature for 1 h to passivate the reduced catalysts. In this way, the passivated catalysts can form an oxide layer on the surface of the active metal to prevent excessive oxidation of the active metal, when the catalysts were exposed to air.

The X-ray diffraction (XRD) patterns of catalysts were obtained on X-ray diffractometer (SHIMADZU XRD-6000) at a scanning rate of 5°

min^{-1} from 10 to 90° , which was equipped with Cu K α ($\lambda = 0.154\text{ nm}$) radiation (40 kV, 30 mA). The crystallite phase in the sample was determined by comparing the XRD pattern to a standard XRD pattern of the International Diffraction Data Center.

The specific surface area and total pore volume were determined by automated surface area and pore size analyzer from nitrogen adsorption isotherms measurement, which was carried out at -196°C with a using a Quantachrome automated gas sorption analyzer (Autosorb-iQ-C). Prior to measurement, the sample was degassed at 180°C for 6 h under vacuum.

Temperature-programmed desorption (TPD) was carried out on Quantachrome automated gas sorption analyzer. About 0.1 g amount of sample was pretreated at 200°C for 1 h under N_2 . NH_3 (CO_2) adsorption was performed at room temperature for 30 min by switching Ar flow to a stream of 5 vol % NH_3 (CO_2)-Ar (30 mL min^{-1}). The sample was purged with Ar (30 mL min^{-1}) for 2 h in order to remove the physically adsorbed NH_3 (CO_2). TPD test was carried out under the condition of a constant flow of Ar (30 mL min^{-1}) from 50°C to 600°C at a heating rate of 5°C min^{-1} . The concentration of NH_3 (CO_2) in the tail gas was continuously monitored by a TCD detector.

The spectra of in-situ X-ray photoelectron spectroscopy (XPS) were recorded on SHIMADZU AXIS Supra X-ray photoelectron spectrometer with an Al K α X-ray resource. Before in-situ XPS characterization, the used catalysts were treated in reaction chamber by the syngas ($\text{H}_2/\text{CO} = 1$, GHSV = $9000\text{ mL/g-Fe}_2\text{O}_3\text{ h}^{-1}$) for 2 h at 280°C . The samples were then cooled to room temperature and transferred to the analysis chamber under vacuum for measurement. The peaks were fitted using Gaussian-Lorentzian curves after Shirley or Tougaard background subtraction. All of the binding energies were calibrated by the C1s peak at 284.8 eV .

In order to obtain the morphology, structural and size characteristics of the catalyst, the high-resolution transmission electron microscope (HR-TEM) and scanning electron microscope (SEM) results were performed on JEM-2100 F (JEOL) and Zeiss Supra55, respectively. The HR-TEM specimen was prepared by ultrasonically the sample in ethanol. Afterward, a drop of suspension was deposited on Ultra-thin carbon film grids and dried in air.

The H_2 (CO) temperature programmed reduction (H_2 (CO)-TPR) experiments were performed in a quartz tube fixed-bed reactor, which equipped with a thermal conductivity detector (TCD) to record the H_2 (CO) content of the tail gas. The selected fresh catalysts (20–40 mesh particles) of $\alpha\text{-Fe}_2\text{O}_3$ (0.05 g) together with 0.05 g quartz sand, $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ (0.1 g) and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ (0.1 g), were in-situ pretreated for 1 h at 120°C using high purity Ar stream (30 mL min^{-1}) to remove the excess water. When the temperature cooled down to room temperature, the catalysts were exposed to the mixed gas 10 % H_2 (CO)/90 %Ar (30 mL min^{-1}). Subsequently, the temperature is raised to 800°C at a heating rate of 10°C/min .

Mössbauer spectroscopy experiments were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany) at room temperature. The radioactive source was ^{57}Co source in Pd matrix. Data analysis was performed using a nonlinear least-squares fitting routine that models the spectra as a combination of singlets, quadruple doublets, and magnetic sextuplets based on a Lorentzian line shape profile. The spectral components were identified based on their isomer shift (IS), quadruple splitting (QS), and magnetic hyperfine (Hhf).

The in-situ diffuse reflectance infrared Fourier transform spectra (DRIFTS) were collected on a Bruker TENSOR 27 FT-IR spectrometer, which equipped with mercury cadmium telluride (MCT) detector at the liquid nitrogen temperature and a diffuse reflectance infrared cell with ZnSe window. Before in-situ DRIFTS characterization, the pre-reduced and passivated catalysts were diluted by KBr with catalyst : KBr = 1:30, and then the diluted samples (about 15 mg) could be re-reduced in diffuse reflectance infrared cell by the syngas ($\text{H}_2/\text{CO} = 1$, 30 mL min^{-1}) for 1 h at 350°C and atmospheric pressure. The catalysts were exposed to the CO gas for 1 h at room temperature. The DRIFTS

were obtained after the re-reduced catalysts were exposed the purity N₂ stream (30 mL min⁻¹) for 1 h. All the spectra were recorded at a resolution of 4 cm⁻¹ and accumulation of 32 scans.

2.4. Catalytic reaction

FTS reactions were carried out in fixed-bed reactor (stainless steel, 8 mm (i.d.)). The flow rates of the gases were measured using mass flow meters (5850E, Brooks). The syngas (H₂/CO = 1) contained 5% argon as internal standard. The prepared catalyst was placed in the reactor tube, and the reactor was raised to 350 °C at a heating rate of 2 °C/min at 0.1 MPa. The reduction progress was then carried out in-situ for 10 h by syngas. After that, the reactor cooled down to room temperature. Before the FTS reactions, the pressure of the reactor was regulated to 1.0 MPa by the syngas (H₂/CO = 1, GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹), the temperature of reactor was ramped to 280 °C at a heating rate of 10 °C/min. The catalysts tests were carried out at the operation conditions: 280 °C, 1.0 MPa, H₂/CO = 1, GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹.

The FTS reaction products were analyzed by two GC-2014C gas chromatographs (Shimadzu) using two detectors. The effluent gas (Ar (internal standard), CO, CO₂, CH₄) were analyzed online by thermal conductivity detector (TCD) with activate carbon column. Another flame ionization detector (FID) was used to analyze the hydrocarbons (C₁-C₅) with Porapak-Q column online. The analyses of liquid products were carried out by silicone SE-30 column offline on GC-FID. The mass balance based on the moles of carbon for all reaction was calculated, which was almost 95 %.

3. Results and discussion

3.1. XRD

XRD (X-ray diffraction) was performed to obtain the crystal structure of catalysts. As shown in Figure S1, all of fresh catalysts showed the diffraction peaks in agree with the crystalline α -Fe₂O₃ phase (PDF#33-0664). In addition, for the SiO₂/ α -Fe₂O₃ = 1 catalyst and Al₂O₃/ α -Fe₂O₃ = 1 catalyst, we could clearly see the diffraction peaks ascribed to Al₂O₃ phase (PDF#10-0173) and a broad peak appeared at around 25.0° (amorphous SiO₂), respectively. In contrast, the diffraction peaks of α -Fe₂O₃ phase were hardly observed over the reduced catalysts (Fig. 1a), which was quite consistent with the XRD patterns of used catalysts (Fig. 1b), indicating the most of iron oxide were reduced under reduction progress. Meanwhile, the XRD patterns of reduced and used catalysts show that the diffraction peaks associated with the iron carbide phases at around $2\theta = 40^\circ - 50^\circ$, where exist main peaks of several iron carbides species such as Fe₃C, Fe₅C₂ and Fe₂C. As always recognized as the active phase for FTS reaction [27,28], different types of iron carbides are formed by carburization during reduction progress and FTS reaction [16,29,30]. However, it is worth noting that to obtain further information about the iron carbide phases by XRD patterns is challenging due to very similar peak location. On the other hand, the XRD patterns of the reduced and used samples are very similar, including intensity and location of diffraction peaks, indicating the similar composition of iron carbides for the reduced samples and the used samples in this study. Besides, the diffraction peaks assigned to Al₂O₃ and SiO₂ were also observed for the reduced and used catalysts, which implied the physical mixture structure of catalysts was barely destroyed during the reduction progress and FTS reaction.

3.2. XPS and texture structure

As illustrated in Figure S2, the XPS (X-ray photoelectron spectroscopy) was carried out to further understand electronic state of Fe atoms on the surface of various fresh catalysts. The spectra of Fe 2p show that the Fe 2p_{1/2} and Fe 2p_{3/2} peaks are observed at around 724.1 eV and 710.8 eV, and a very weak satellite can be discovered

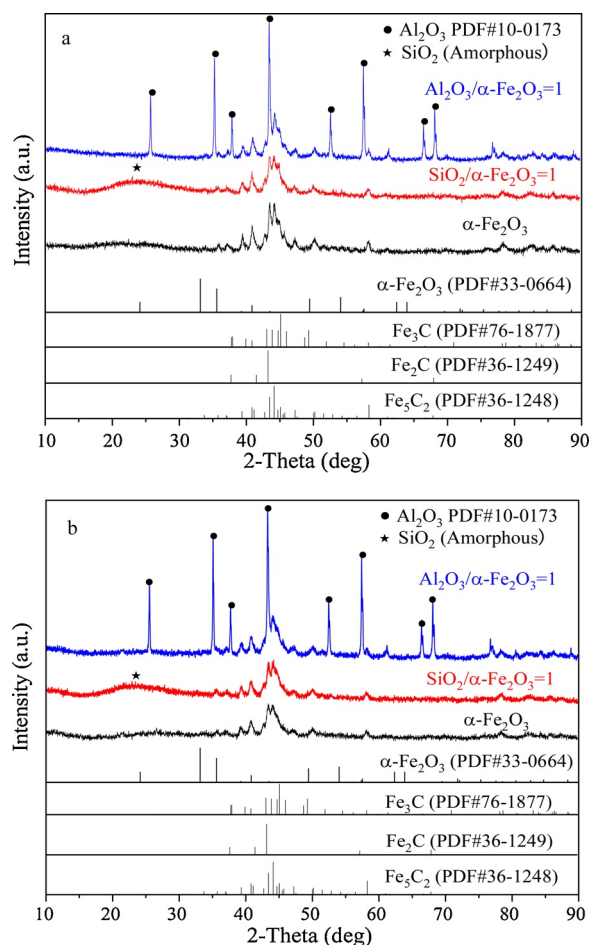


Fig. 1. The XRD patterns of reduced (a) and used (b) catalysts. Reduction conditions: 350 °C, 0.1 MPa, H₂/CO = 1, GHSV = 9000 mL/g-Fe₂O₃ h⁻¹; Reaction conditions: 280 °C, 1.0 MPa, H₂/CO = 1, GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹.

between them, which demonstrates the iron species on the surface of catalysts is typical α -Fe₂O₃ phase [15,31], which is consistent with the results of XRD. As the three samples exhibits almost the same XPS curves, it is believed that the physical mixture of fresh iron oxide powder with silica or alumina powder do not influence the properties of surface iron atoms for fresh catalysts, contributing to clearly investigating the formation and effects of different iron carbides.

N₂ adsorption-desorption isotherms show representative type-II curves (Figure S3), which is normally considered to be responsible for nonporous adsorbents [8,15,32]. The BET surface area, and pore volume of pure α -Fe₂O₃ are calculated to be 6.72 m² g⁻¹ and 0.019 cm³ g⁻¹, respectively. These results indicate that α -Fe₂O₃ powder is a nonporous structure, which could avoid the influence of diffusion limitation on the catalytic performance during the FTS reaction, contributing to revealing the nature of different catalysts.

3.3. SEM images

The scanning electron microscopy (SEM) images (Fig. 2) of fresh catalysts indicates that the particle size of α -Fe₂O₃ powders is above 300 nm, which can avoid the formation of the packed micro pores between the α -Fe₂O₃ particles, contributing to good diffusion rate of reactants and products during the FTS reaction. The pure SiO₂ and Al₂O₃ powders also show the similar particle size to α -Fe₂O₃ powders, contributing to good diffusion rate of the mixed catalysts. In addition, the EDX elemental mapping of SiO₂/ α -Fe₂O₃ = 1 catalyst and Al₂O₃/ α -Fe₂O₃ = 1 catalyst is shown in Figure S4, in which Al₂O₃ and SiO₂ are

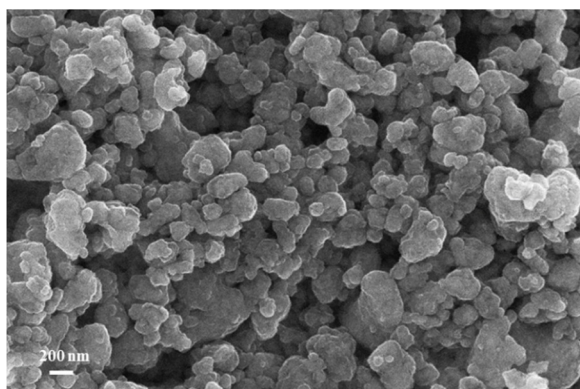


Fig. 2. SEM image of the α -Fe₂O₃ catalyst.

uniformly distributed around α -Fe₂O₃, indicating the powders were well mixed.

3.4. TEM images

The details of the formed iron carbides of various used catalysts were investigated by HR-TEM. The TEM image of the used pure α -Fe₂O₃ catalyst is shown in Fig. 3a. It can be seen in Fig. 3a that χ -Fe₅C₂ and θ -Fe₃C carbides co-exist on the pure α -Fe₂O₃ catalyst. However, for the mixed catalysts, SiO₂/ α -Fe₂O₃ = 1 catalyst and Al₂O₃/ α -Fe₂O₃ = 1 catalyst, the ϵ -Fe₂C species clearly appear besides the χ -Fe₅C₂ and θ -Fe₃C species. More important, the HR-TEM images (Fig. 3) shows that the ϵ -Fe₂C was formed on the interfaces of α -Fe₂O₃-SiO₂ and α -Fe₂O₃-Al₂O₃, respectively, indicating that the mixed SiO₂ and Al₂O₃ powders influenced the formation iron carbides contributing to forming the ϵ -Fe₂C species. And above results strongly suggest that the mixture of SiO₂ and Al₂O₃ powders is vital for the formation of C-rich iron carbide ϵ -Fe₂C species on the iron-based catalyst.

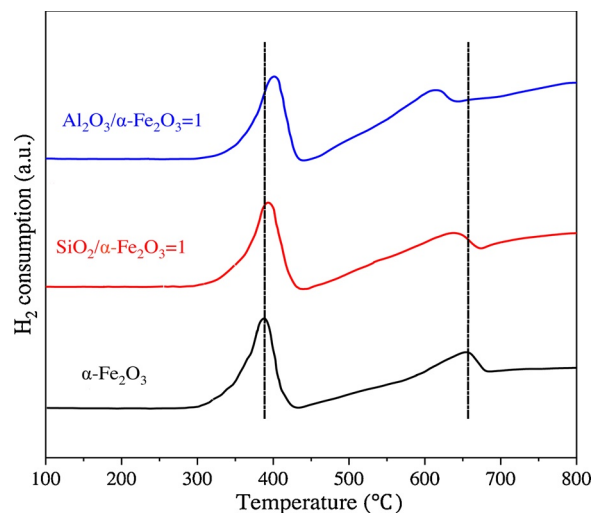


Fig. 4. H₂-TPR profiles of various fresh catalysts.

3.5. H₂-TPR and CO-TPR

The reduction behavior of various fresh catalysts was characterized by H₂ temperature programmed reduction (H₂-TPR) to evaluate the influence of the SiO₂ and Al₂O₃. As given in Fig. 4, it can be clearly seen all the profiles of the fresh catalysts have two similar main reduction peaks, which are well known that the Fe-based catalysts occurs in two steps in the reduction process. The first peak is corresponding to the reduction from α -Fe₂O₃ → Fe₃O₄ at the low temperature, while the second peak is ascribed to the reduction of Fe₃O₄ → FeO and/or Fe at the high temperature [15,33,34]. Moreover, it can be obviously found that the profiles of the SiO₂/ α -Fe₂O₃ = 1 catalyst and the Al₂O₃/ α -Fe₂O₃ = 1 catalyst slight shifted to higher temperature compared with that of α -Fe₂O₃ catalyst at first reduction step. These results show that the first reduction step of the fresh catalysts is slight suppressed by the

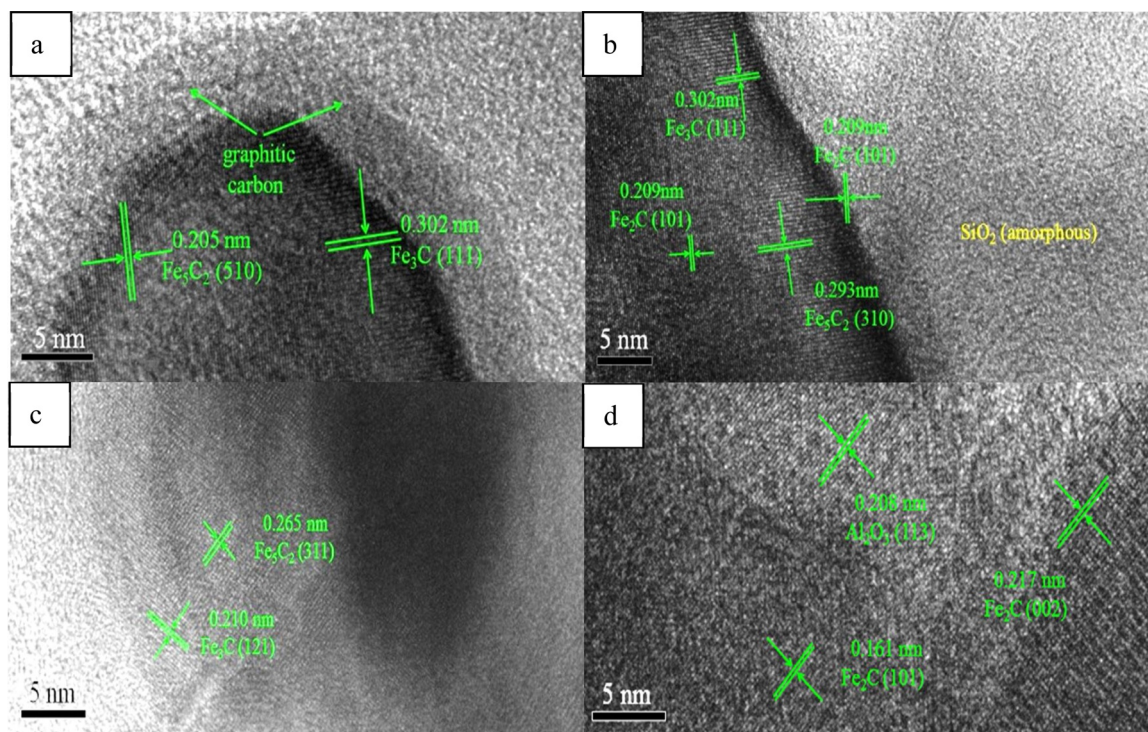


Fig. 3. HR-TEM of catalysts after FTS reaction: a) α -Fe₂O₃ catalyst, b) SiO₂/ α -Fe₂O₃ = 1 catalyst, c) and d) Al₂O₃/ α -Fe₂O₃ = 1 catalyst. Reaction conditions: 280 °C, 1.0 MPa, H₂/CO = 1, GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹.

introduction of SiO_2 and Al_2O_3 due to the formed interface between the $\alpha\text{-Fe}_2\text{O}_3$ and the added oxide, as illustrated by TEM images. On the contrary, it is evident from the second reduction step that this inhibitive effect of SiO_2 and Al_2O_3 are obviously reversed on the higher temperature reduction peaks. The second reduction peak gradually shifts to lower temperature with the addition of SiO_2 and Al_2O_3 , indicating the interaction between the added oxide and $\alpha\text{-Fe}_2\text{O}_3$ is favorable for further reduction of iron oxide at high temperature. In fact, Ding et al. successfully proved that carburization ability is following as $\alpha\text{-Fe} > \text{FeO} > \text{Fe}_3\text{O}_4 > \alpha\text{-Fe}_2\text{O}_3$ [35]. Therefore, for the mixed catalyst the lower reduction temperature of $\text{Fe}_3\text{O}_4 \rightarrow \text{FeO}$ and/or Fe would be advantageous to carburization and forming C-rich iron carbide phase, as confirmed by the HR-TEM results. On the other hand, the distinct reducibility of catalysts indicates that the Fe-Si and Fe-Al interactions were formed on the surface of iron oxide in $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst during the reduction process, respectively. Therefore, it is believed that the electronic state of Fe atoms on the surface of iron oxide has been influenced due to the incorporation of SiO_2 and Al_2O_3 , which may affect the formation of iron carbide phases on the surface of catalyst during the reduction. Meanwhile, the lowest reduction temperature of the $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst on the second reduction step implies that the Fe-Al interaction could facilitate to form more C-rich iron carbide phase ($\epsilon\text{-Fe}_2\text{C}$). In order to further to understand the carburization behavior of various fresh catalysts, the CO-TPR of various catalysts was illustrated in Figure S6. Similar with the H_2 -TPR two reduction steps and conclusion, the carburization process of $\alpha\text{-Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4$ ($\sim 350^\circ\text{C}$) $\rightarrow \text{FeO}$ and/or Fe ($400^\circ\text{C} \sim 600^\circ\text{C}$) [36–38] was also observed on the three catalysts. Significantly, it is worth noting that the peak appeared at the 630°C can be assigned to the carburization process of $\text{Fe} \rightarrow \text{Fe}_3\text{C}$ [39,40]. Moreover, the lowest carburization temperature and the highest CO consumption of the $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst on the three carburization steps imply that the Fe-Al interaction could facilitate to form more C-rich iron carbide phase ($\epsilon\text{-Fe}_2\text{C}$), which is agree well with the analysis of H_2 -TPR results.

3.6. In-situ XPS

It is well known that the performance of catalyst is associated with the iron carbide phases on the catalyst surface during the FTS reaction [8–10,13,15,41]. In order to discover the effects of the mixed SiO_2 and Al_2O_3 on the formation of iron carbides, the in-situ XPS was applied to analyze the different phases of iron carbide on the catalyst surface. Before in-situ XPS characterization, the used catalysts were treated in reaction chamber by the syngas ($\text{H}_2/\text{CO} = 1$, GHSV = 9000 mL/g- $\text{Fe}_2\text{O}_3 \text{ h}^{-1}$) for 2 h at 280°C . The spectra of in-situ XPS are shown in Fig. 5, and the binding energies of different iron species are listed in Table S1. After deconvolution, notable difference of the Fe 2p XPS spectra proves that several different iron carbide species were formed on the surface of the catalysts. For the pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst, peaks at 707.0 eV and 719.9 eV are assigned to $\text{Fe}2\text{p}_{3/2}$ and $\text{Fe}2\text{p}_{1/2}$ of $\chi\text{-Fe}_5\text{C}_2$ species [14,42], respectively, and peaks at 707.9 eV and 720.6 eV should be assigned to the $\theta\text{-Fe}_3\text{C}$ species [43,44]. Meanwhile, two very weak peaks appear at 706.1 eV and 718.8 eV, which should be the $\epsilon\text{-Fe}_2\text{C}$ species [45]. However, for the $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst, the peaks of $\epsilon\text{-Fe}_2\text{C}$ species obviously become stronger than that of pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst, and the content of $\epsilon\text{-Fe}_2\text{C}$ phase of $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst is the highest within three samples as compared in Table S1, indicating the mixed Al_2O_3 effectively improves the formation of $\epsilon\text{-Fe}_2\text{C}$ species on the $\alpha\text{-Fe}_2\text{O}_3$. Meanwhile, the content of $\theta\text{-Fe}_3\text{C}$ species decreased obviously for two mixture catalysts. On the other hand, for the $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst, it is clearly shown that all the iron carbide phases gradually shift to higher binding energies than that of $\alpha\text{-Fe}_2\text{O}_3$ catalyst in this study, indicating that the electron transfer between Fe and Al or Si was enhanced, which is in line with the first reduction peak in H_2 -TPR. As previously reported [15], for the reduced $\gamma\text{-Fe}_2\text{O}_3$ nanorod

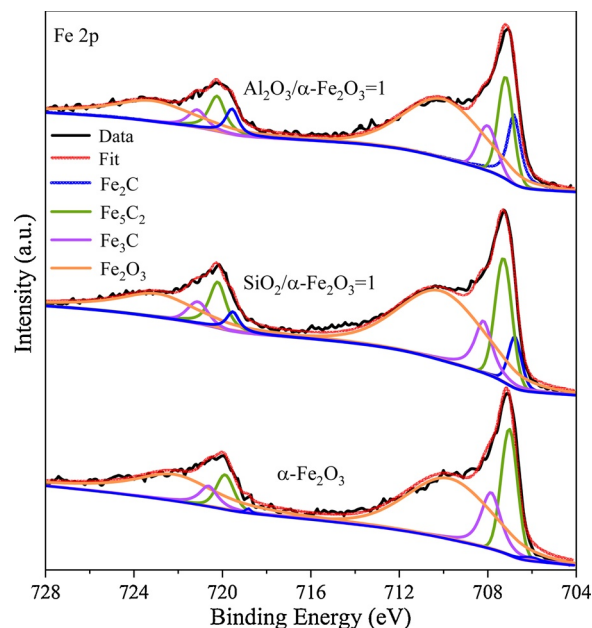


Fig. 5. In-situ XPS spectra of Fe 2p for various used catalysts.

catalyst, the higher binding energy of iron atom contributed to forming more C-rich iron carbide $\chi\text{-Fe}_5\text{C}_2$. Therefore, it is believed that the higher BE of iron species, which induced by the mixed SiO_2 and Al_2O_3 , are advantageous to forming more C-rich iron carbide $\epsilon\text{-Fe}_2\text{C}$. Combining the HR-TEM results, it is reasonable to believe that the mixed SiO_2 and Al_2O_3 have a significance effect on the electronic state of Fe atoms, resulting in forming more $\epsilon\text{-Fe}_2\text{C}$ on the surface of $\alpha\text{-Fe}_2\text{O}_3$ catalysts.

3.7. Mössbauer spectroscopy

Using Mössbauer spectroscopy at room temperature, it was possible to ascertain accurate values and composition of the iron-based materials, especially for iron carbides [46]. Therefore, in order to obtain the accurate values and composition of different iron species, the used catalysts, after 30 h reaction, were investigated in detail via Mössbauer spectroscopy. As shown in Fig. 6, the Mössbauer spectra of all used catalysts shows the coexistence of $\chi\text{-Fe}_5\text{C}_2$, $\theta\text{-Fe}_3\text{C}$ and $\epsilon\text{-Fe}_2\text{C}$, and the mixed catalysts exhibit more strong fitting peaks of $\epsilon\text{-Fe}_2\text{C}$, which is in good accordance with the analysis of in-situ XPS. As summarized in Table 1, the results of Mössbauer spectra show that the $\chi\text{-Fe}_5\text{C}_2$, with similar content for all used catalyst, is the mainly iron carbide phase, which has already recognized as the active phase for the FTS reaction [12,47,48]. In addition, the oxide (SiO_2 , Al_2O_3) mixed $\alpha\text{-Fe}_2\text{O}_3$ catalysts exhibit higher total iron carbides content than the pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst after FTS reaction, indicating the mixed SiO_2 or Al_2O_3 truly improved the carburization of iron. Furthermore, the C-rich iron carbide phase ($\epsilon\text{-Fe}_2\text{C}$) was as low as 0.6 % in pure $\alpha\text{-Fe}_2\text{O}_3$ catalyst. However, for the used catalysts of the $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst, the content of C-rich iron carbide phase $\epsilon\text{-Fe}_2\text{C}$ significantly increases to 9.8 % and 14.2 %, respectively. It is proposed that the more C-rich iron carbide phase $\epsilon\text{-Fe}_2\text{C}$ could be formed on the surface of $\alpha\text{-Fe}_2\text{O}_3$ due to higher carburization ability for the presence of SiO_2 and Al_2O_3 , as confirmed by in-situ XPS and second reduction peak in H_2 -TPR. Meanwhile, the content of the $\theta\text{-Fe}_3\text{C}$ markedly decreases for two mixed catalysts. As the different iron carbide phase has different hydrogenation ability [8,15,49], it is believed that the decreased $\theta\text{-Fe}_3\text{C}$ content and the increased $\epsilon\text{-Fe}_2\text{C}$ content would contribute to different reaction performance of FTS for the mixed catalysts.

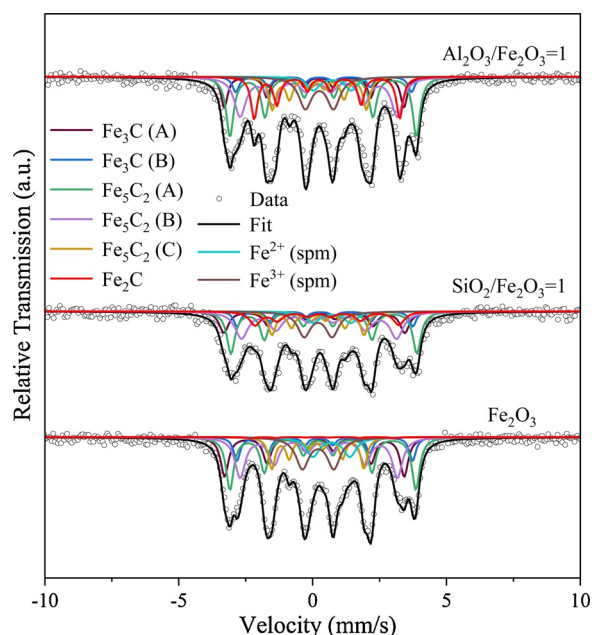


Fig. 6. Mössbauer spectra of various used catalysts after FTS reaction. Reaction conditions: 280 °C, 1.0 MPa, $H_2/CO = 1$, GHSV = 10,000 mL/g- Fe_2O_3 h $^{-1}$.

3.8. In-situ CO drifts

In-situ diffuse reflectance infrared Fourier transform spectra (DRIFTS) experiments were carried out to further understand the behavior of CO adsorption on iron carbide phases at room temperature. The adsorption of CO on the various catalysts was compared at Fig. 7. It is reported that the adsorbed CO bands on the iron carbides are appeared at the range of 2050–1900 cm^{-1} [50], in which the adsorbed CO bands above 2000 cm^{-1} is recognized as the linear adsorption of CO on the surface of catalyst [51,52]. In case of present results, it can be seen that the linear adsorption peak intensity of CO increases in order of α - Fe_2O_3 < SiO_2/α - Fe_2O_3 < Al_2O_3/α - Fe_2O_3 , which is the same order to

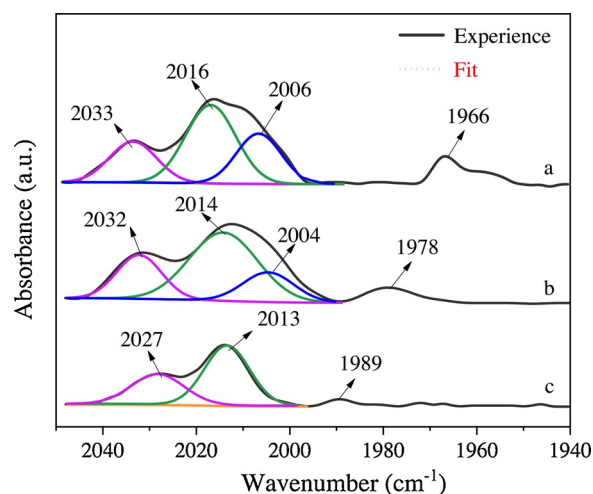


Fig. 7. In-situ CO-DRIFT spectra of the reduced catalysts. a) Al_2O_3/α - $Fe_2O_3 = 1$; b) SiO_2/α - $Fe_2O_3 = 1$; c) α - Fe_2O_3 .

the content of iron carbide, confirmed by the in situ XPS and Mössbauer spectroscopy. On the other hand, the strong CO adsorption indicates the coverage of CO on the catalysts surface is enhanced, which in turn increases the possibility of promoting FTS activity, suggesting the higher catalytic activity might be achieved on the Al_2O_3/α - $Fe_2O_3 = 1$ catalyst. Furthermore, based on the BE value of different iron carbides from in situ XPS, the electron density of θ - Fe_3C phases is lower than that of χ - Fe_5C_2 species. As the increased charge transfer from active phase to CO could let CO bond become weaker [53], the CO linear adsorption appears at around 2030 cm^{-1} and 1915 cm^{-1} should be attributed to the CO adsorption on θ - Fe_3C and χ - Fe_5C_2 phases, respectively. In contrast, the additional CO linear adsorption of SiO_2/α - Fe_2O_3 and Al_2O_3/α - Fe_2O_3 catalysts at the 2004 cm^{-1} and 2006 cm^{-1} is obviously different from the α - Fe_2O_3 catalyst, which should be assigned to the CO linear adsorption of the ϵ - Fe_2C phase. Significantly, after fitted the curve of the CO linear adsorption, it is found that the ratio of different CO linear adsorption (Table S2) are highly dependent on the content of various

Table 1

Mössbauer parameters of the used catalysts. Reaction conditions: 280 °C, 1.0 MPa, $H_2/CO = 1$, GHSV = 10,000 mL/g- Fe_2O_3 h $^{-1}$.

Catalysts	Phases	Mössbauer parameters			
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area (%)
α - Fe_2O_3	Fe_2C	0.31	0.18	167	0.6
		0.32	0.18	216	22.4
		0.27	−0.04	183	22.4
		0.18	0.07	107	11.2
	Fe_3C	0.18	−0.18	209	17.1
		0.37	0.22	204	8.6
		0.72	1.38		6.8
		0.23	1.17		10.9
SiO_2/α - $Fe_2O_3 = 1$	Fe_2C	0.40	0.30	167	9.8
		0.33	0.21	216	22.2
		0.28	−0.07	180	22.2
		0.22	−0.01	107	11.2
	Fe_3C	0.19	−0.23	210	12.1
		0.39	0.18	204	6.0
		0.74	1.39		5.8
		0.22	1.05		10.7
Al_2O_3/α - $Fe_2O_3 = 1$	Fe_2C	0.40	0.30	169	14.2
		0.33	0.16	216	22.1
		0.23	0.01	182	22.1
		0.22	0.10	109	11.2
	Fe_3C	0.16	−0.17	209	11.4
		0.34	0.16	204	5.7
		0.75	1.42		3.2
		0.27	1.05		10.1

iron carbides from Mössbauer spectroscopy and in-situ XPS. On the other hand, compare with the χ -Fe₅C₂ and θ -Fe₃C, the red shifted linear adsorbed CO on the ϵ -Fe₂C phase, indicating that the C–O bond is more weakened, is more easily to be active and dissociated on surface of catalyst. Hence, it is reasonable to conclude that the ϵ -Fe₂C phase could be more active in enhancing the catalytic activity.

The spectra below 2000 cm⁻¹ can be ascribed to the bridge-type adsorption of CO [54,55]. The bridge-type CO is a weaker CO bond and more easily dissociated into carbon and oxygen, accounting for the excellent catalytic performance toward FTS activity [54]. The stronger peak of bridge-type CO adsorption is identified on the Al₂O₃/α-Fe₂O₃ = 1 catalyst and SiO₂/α-Fe₂O₃ = 1 catalyst compared with α-Fe₂O₃ catalyst, implying the higher catalytic activity might be achieved on the SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst, especially for the Al₂O₃/α-Fe₂O₃ = 1 catalyst. On the other hand, the stronger adsorption and red shift of CO bridge-type adsorption further suggest that on the surface of the SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst the existence of ϵ -Fe₂C species would facilitate the formation of highly active adsorbed CO species, including linear and bridge adsorbed CO, contributing to highly catalytic activity during the FTS reaction.

3.9. FTS reaction performance

The catalytic performance of various catalysts in the FTS reaction was tested at 280 °C, 1 MPa, H₂/CO = 1, and GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹ for 30 h. As shown in Fig. 8 and Table 2, SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst show a very short induction period and reach a steady state from the second hour. Meanwhile, they realize excellent stability as without any deactivation within 30 h. It is believed that the active phase of these two catalysts is stable after carburization by reduction and during FTS reaction, as proved by XRD results of the reduced and used catalyst. In contrast, an obvious deactivation was observed after reach the maximum activity after 3 h reaction for the pure α-Fe₂O₃ catalyst. The HR-TEM images of used α-Fe₂O₃ catalyst clearly show that the iron carbide phases are covered by graphitic carbon layer over the used α-Fe₂O₃ catalyst (Fig. 3), which is not observed on the used SiO₂/α-Fe₂O₃ = 1 and Al₂O₃/α-Fe₂O₃ = 1 catalyst. The TEM images of the reduced and used α-Fe₂O₃ catalyst are shown in Figure S5. It is noted that the crystalline size of pure α-Fe₂O₃ catalyst is not obviously changed during the FTS reaction, which is agree well with the XRD results (Fig. 1). However, the only clearly change of the reduced and used α-Fe₂O₃ catalyst is the formation of graphitic carbon layer on the used α-Fe₂O₃ catalyst, which is recognized as the reason for the deactivation. Meanwhile, at steady state

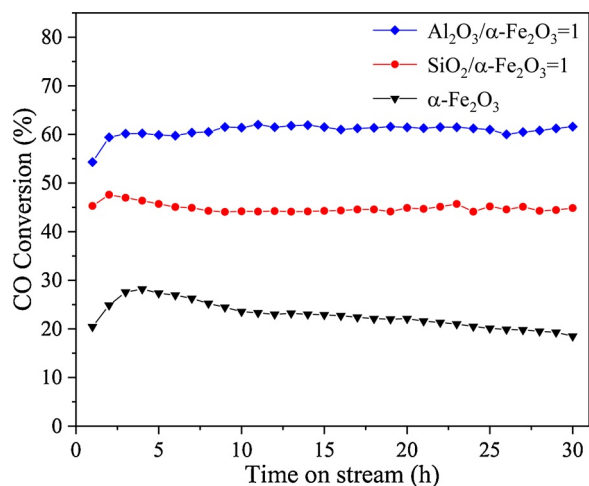


Fig. 8. CO conversion and stability test of the catalysts. Reaction conditions: 280 °C, 1.0 MPa, H₂/CO = 1, W/F(CO + H₂ + Ar) = 5 g-cat·h·mol⁻¹.

of FTS reaction, the Al₂O₃/α-Fe₂O₃ = 1 catalyst shows the highest 61.6 % CO conversion, which is 3.3 times higher than that of the pure α-Fe₂O₃ catalyst (18.5 %). Besides, SiO₂/α-Fe₂O₃ = 1 catalyst realizes as high as 44.9 % CO conversion. On the other hand, as compared in Table 2, the selectivity of light olefins (C₂-C₄) is decreasing with the increased CO conversion for three catalysts, and the olefin/paraffin ratio (O/P) in light olefins has same tendency with the selectivity of light olefin. As the decreased θ -Fe₃C content, as proved by in-situ XPS and Mössbauer spectroscopy, results in poor light olefins selectivity for two mixed catalysts, it is considered the θ -Fe₃C is in favor of forming light olefins in FTS reaction [8,15]. As all used catalysts have similar content of χ -Fe₅C₂ based on results of Mössbauer spectroscopy, these observed results strongly suggest that the FTS activity is enhanced by the existence of ϵ -Fe₂C active phase, and the catalytic activity (FTY in Table 2) has positive relationship with the increased content of ϵ -Fe₂C as predicted by the theoretical calculation of density functional theory (DFT) [49]. Therefore, we suggest that the excellent catalytic activity of the mixed catalysts, especially for Al₂O₃/α-Fe₂O₃ = 1 catalyst, is realized due to the improved formation of ϵ -Fe₂C active phase. However, in Table 2, the poor catalytic performance was achieved for the calcined SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst (calcined at 600 °C) under the same FTS reaction conditions, which can be attributed to the strong interaction of Fe-Si and Fe-Al, respectively.

3.10. Discussion

In this study, the designed nonporous structure of α-Fe₂O₃ catalysts successfully avoided the influence of pore size, iron particle size, dispersion, metal/promoter-inert support interaction, which are acknowledged as an ideal choice to reveal the exact role of each iron carbide in the activity during the FTS. On the other hand, the nonporous structure of α-Fe₂O₃ powders makes the active phase iron carbides concentrated on the outer surface of the catalyst, facilitating to probe the surface property evolution of Fe atoms during FTS reaction by various characterizations including the perfect in-situ XPS spectra and CO-DRIFTS.

From the XRD patterns in Fig. 1, it can be found that the peak intensity and location of the iron carbides is almost the same for the reduced and the used mixed catalyst, indicating that the iron carbides species are not significantly dynamic change during the FTS reaction due to the special mixed powders structure. The TEM images (Fig. 3) and H₂-TPR (Fig. 4) results reveal that the Fe-Si and Fe-Al interaction were formed on the surface of SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst during the FTS reaction, respectively. Meanwhile, the mixed oxides show critical effect on the formation of different iron carbides via altering electronic state of Fe atoms on the catalysts surface. Meanwhile, the in-situ XPS (Fig. 5) and Mössbauer spectra (Fig. 6) results confirmed that the C-rich ϵ -Fe₂C phase was formed on the SiO₂/α-Fe₂O₃ = 1 catalyst and Al₂O₃/α-Fe₂O₃ = 1 catalyst surface. On the other hand, in Fig. 7, the in-situ CO-DRIFTS clearly shows the linear adsorbed CO species on the different iron carbides, proving the results of in-situ XPS and Mössbauer spectra, indicating the ϵ -Fe₂C phase is more active.

As illustrated in Fig. 9, the CO conversion of various catalysts is correlated with the content of different iron carbides. It can be seen from Fig. 9 that the content of ϵ -Fe₂C increases in order of α-Fe₂O₃ < SiO₂/α-Fe₂O₃ < Al₂O₃/α-Fe₂O₃ based on Mössbauer parameters (Fig. 9a), in-situ XPS data (Fig. 9b) and in-situ CO-DRIFT data (Fig. 9c), which is the same order to CO conversion. For the χ -Fe₅C₂, which has been recognized as active sites of iron catalysts in FTS reaction for several decades [11–14], its content is very similar for all samples as shown in Fig. 9 a, and decreases in order of α-Fe₂O₃ > SiO₂/α-Fe₂O₃ > Al₂O₃/α-Fe₂O₃ based on in-situ XPS data. Meanwhile, the pure α-Fe₂O₃ catalyst contains the highest amount of θ -Fe₃C in three catalysts, and SiO₂/α-Fe₂O₃ and Al₂O₃/α-Fe₂O₃ catalysts have almost the same content of θ -Fe₃C based on both of Mössbauer parameters and in-

Table 2
The activity and hydrocarbon selectivity of different iron-based catalysts ^a.

Catal.	CO conv. (%)	CO ₂ sel. (%)	Hydrocarbon Selectivity (c-mol %)				O/P ^b	α^c	FTY ^d 10 ⁻⁵ mol CO ^g Fe ₂ O ₃ ⁻¹ s ⁻¹
			CH ₄	C ₂ -C ₄ Olefins	C ₂ -C ₄ Paraffins	C ₅ +			
α -Fe ₂ O ₃	18.5	35.9	17.1	40.6	9.6	32.7	4.2	0.71	1.56
SiO ₂ / α -Fe ₂ O ₃ = 1	44.9	25.1	18.7	36.8	11.0	33.5	3.3	0.73	3.78
Al ₂ O ₃ / α -Fe ₂ O ₃ = 1	61.6	33.9	16.0	31.2	16.1	36.7	1.9	0.76	5.18
SiO ₂ / α -Fe ₂ O ₃ = 1 ^e	31.1	30.3	16.9	33.6	11.7	37.8	2.8	0.77	2.62
Al ₂ O ₃ / α -Fe ₂ O ₃ = 1 ^e	38.6	29.7	16.1	30.1	14.6	49.2	2.1	0.78	3.25

^a Reaction conditions: 280 °C, 1.0 MPa, H₂/CO = 1, GHSV = 10,000 mL/g-Fe₂O₃ h⁻¹.

^b olefin to paraffin in the C₂-C₄ range hydrocarbons (c-mol %).

^c Chain growth probability.

^d Calculated by FTY = moles of CO converted per gram of Fe₂O₃ per second.

^e The prepared samples were calcined at 600 °C for 3 h before FTS reaction.

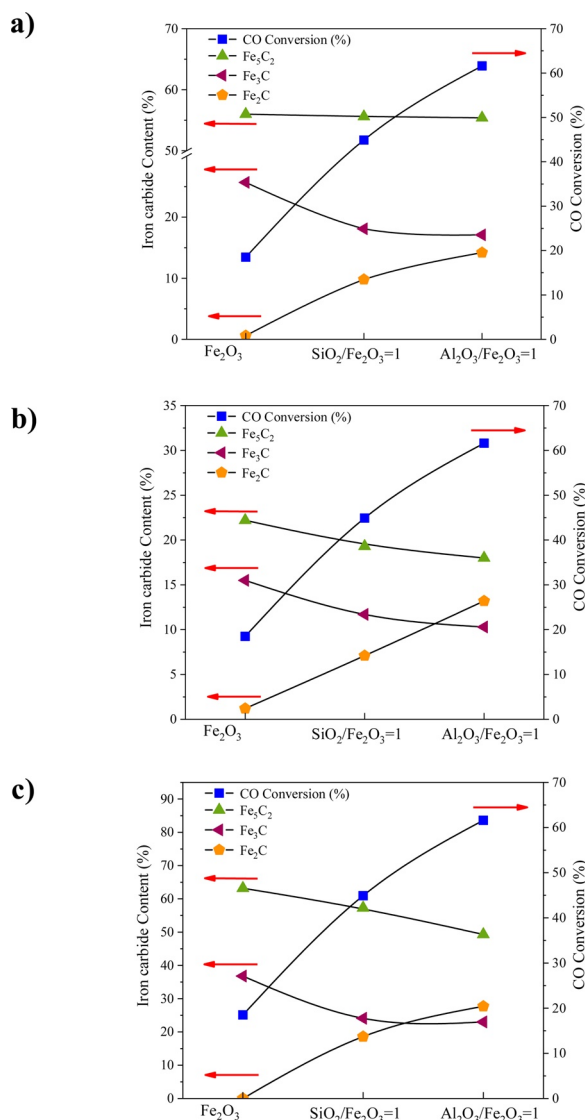


Fig. 9. CO Conversion and iron carbide content as a function of various iron-based catalysts. a) Based on Mössbauer parameters; b) Based on in-situ XPS data; c) Based on in-situ CO-DRIFT data.

situ XPS data. Concerning the CO conversion of various catalysts, the pure α -Fe₂O₃ catalyst shows the lowest CO conversion with the highest content of θ -Fe₃C, indicating the θ -Fe₃C is a lower active phase due to its lower hydrogenation ability [8,15]. Meanwhile, the Al₂O₃/ α -Fe₂O₃ catalyst realizes the highest CO conversion, which is 3.3 times higher

than that of pure α -Fe₂O₃ catalyst, with the highest content of ϵ -Fe₂C and very similar θ -Fe₃C amount to SiO₂/ α -Fe₂O₃ catalyst. The CO-DRIFTS results prove that the linear CO adsorption of ϵ -Fe₂C occurs red shift comparing with χ -Fe₅C₂, indicating the more active the linear CO adsorbed species are formed on the ϵ -Fe₂C. Furthermore, the results of CO-DRIFTS clearly show that the amount of CO adsorption on catalysts containing more ϵ -Fe₂C are much more than pure Fe₂O₃ catalyst. Most importantly, the bridge CO adsorption of the Al₂O₃ mixed catalysts locate at much lower wavenumber, which indicates more active adsorbed CO is formed on the Al₂O₃ mixed catalysts, contributing to the highest activity of this catalyst. Meanwhile, because the catalysts are nonporous materials and have very small specific area as determined by N₂ adsorption isotherms measurement, it can be ignored for the influence on CO conversion induced by diffusion limitation. Therefore, it is considered that the activity of iron-based FTS catalyst, FTY compared in Table 2, has strong positive relationship with the content of ϵ -Fe₂C species, namely the small amount of ϵ -Fe₂C species can significantly improve the activity of iron-based FTS catalyst, comparing to the χ -Fe₅C₂. Significantly, the pure α -Fe₂O₃ catalyst exhibit the relatively high value of olefin/paraffin ratio (O/P = 4.2) in C₂-C₄ fraction, which can be attributed to the nonporous structure reducing the secondary hydrogenation of the formed light olefins, promoting the selectivity of light olefins and the value of O/P. The highest amount of θ -Fe₃C obtained on the pure α -Fe₂O₃ catalyst, which recognized as the effective active phase for the formation of light olefins [8,15], is considered another main reason for the relatively high value of O/P.

In order to verify the Al₂O₃ activation performance for CO, the CO-DRIFT of pure Al₂O₃ was illustrated in Figure S7. When the CO exposed to the Al₂O₃ sample, the two peaks located at 2116 cm⁻¹ and 2171 cm⁻¹ can be attributed to the gas CO. However, almost no linear and bridge adsorption peaks of CO are observed on the CO-DRIFT spectrum at 2050 cm⁻¹ -2000 cm⁻¹ [51,52] and 1800 cm⁻¹ -2000 cm⁻¹ [54,55], respectively. It is indicated that the CO cannot be adsorbed and activated for the Al₂O₃, which strongly suggests that the Al₂O₃ is inert oxide in FTS reaction. Moreover, the Al₂O₃ as an inert oxide support of FTS catalysts has been studied by many other researchers [10,56,57]. Moreover, the acidity and basicity of the SiO₂ and Al₂O₃ were determined by the temperature-programmed desorption (TPD) method using NH₃ and CO₂ as a probing molecule, and the desorption profiles are recorded as shown in Figure S8. Similar NH₃ and CO₂ desorption behaviors could be found for the SiO₂ and Al₂O₃. It is note that the surface acidity and basicity of SiO₂ and Al₂O₃ were very weak. By carefully observation, the two very weak desorption zones can be found. The one below 200 °C represented the NH₃ and CO₂ desorbed from weak acid and basic sites, while the other one occurring at about 300 °C could be assigned to the desorption of NH₃ and CO₂ from strong acid and basic sites. Meanwhile, by carefully observation, the surface acidity and basicity of SiO₂ slightly stronger than Al₂O₃, because the applied Al₂O₃ powder in this study is α -Al₂O₃ which is inactive and

inert. It is well known that the SiO_2 is recognized as the inert oxide support in FTS [58–60], and the extremely low surface acidity and basicity of SiO_2 can be ignored in the catalytic performance and the formation of iron carbides. Hence, it is reasonable for us to further believe that the Al_2O_3 is inert oxide support in FTS, and the extremely low surface acidity and basicity of Al_2O_3 can be ignored in the formation of iron carbides.

On the other hand, although the SiO_2 and Al_2O_3 always are applied as support or structure promoters in iron-based catalyst [2,6,13,16,21,23,25], their effects on the formation of different iron carbides have not been clearly identified, because many factors of catalysts disturb to clarify structure-performance relationship such as surface properties, pore structure, interaction with the supported iron and effects of various promoters. However, in this study, we used nonporous materials successfully avoided complex interplay among many factors, clarifying that the SiO_2 or Al_2O_3 , especially Al_2O_3 , would improve the formation more active $\varepsilon\text{-Fe}_2\text{C}$ species on iron-based catalysts contributing to realizing higher CO conversion in FTS reaction. It is clearly shown in results of in-situ XPS that all the iron carbide phases of the mixed catalysts especially Al_2O_3 , gradually shift to higher binding energies than that of $\alpha\text{-Fe}_2\text{O}_3$ catalyst in this study, indicating that the electron transfer between Fe and Al or Si was enhanced, which is in line with the first reduction peak in H_2 -TPR and CO-TPR. As previously reported [15], for the reduced $\gamma\text{-Fe}_2\text{O}_3$ nanorod catalyst, the higher binding energy of iron atom contributed to forming more C-rich iron carbide $\chi\text{-Fe}_5\text{C}_2$. Therefore, it is believed that the higher BE of iron species, which induced by the mixed SiO_2 and Al_2O_3 , especially Al_2O_3 , are advantageous to forming more C-rich iron carbide $\varepsilon\text{-Fe}_2\text{C}$. Combining the HR-TEM results, it is reasonable to believe that the mixed SiO_2 and Al_2O_3 have a significance effect on the electronic state of Fe atoms, resulting in forming more $\varepsilon\text{-Fe}_2\text{C}$ on the surface of $\alpha\text{-Fe}_2\text{O}_3$ catalysts.

4. Conclusions

In this work, we use physical approach to investigate the new roles of SiO_2 and Al_2O_3 as promoters during the FTS reaction on the nonporous iron-based catalyst and clarify the different role of iron carbide phases. Based on the FTS reaction performance, it is been found that the catalyst activity was greatly enhanced for the presence of SiO_2 and Al_2O_3 . Characterization results reveal that C-rich $\varepsilon\text{-Fe}_2\text{C}$ phase was formed on the $\text{SiO}_2/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst and $\text{Al}_2\text{O}_3/\alpha\text{-Fe}_2\text{O}_3 = 1$ catalyst surface due to the introduction of SiO_2 and Al_2O_3 promoters, suggesting the small amount of the $\varepsilon\text{-Fe}_2\text{C}$ is responsible for the highly activity of the catalysts comparing with the $\chi\text{-Fe}_5\text{C}_2$ and $\theta\text{-Fe}_3\text{C}$. In summary, we use a simple approach to evidences that the clear correlations between the catalysis performance and iron carbides, providing a better understanding of the newly role of inert oxide in formation of different iron carbides.

CRedit authorship contribution statement

Fangxu Lu: Investigation, Data curation, Writing - original draft. **Xin Chen:** Investigation, Data curation. **Zhigang Lei:** Investigation, Data curation. **Lixiong Wen:** Supervision. **Yi Zhang:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare no competing financial interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.apcatb.2020.119521>.

References

- [1] H.M.T. Galvis, K.P. de Jong, ACS Catal. 3 (2013) 2130–2149.
- [2] E. Jiménez-Barrera, P. Bazin, C. Lopez-Cartes, F. Romero-Sarria, M. Daturi, J.A. Odriozola, Appl. Catal. B: Environ. 237 (2018) 986–995.
- [3] L.S. Zhong, F. Yu, Y.L. An, Y.H. Zhao, Y.H. Sun, Z.J. Li, T.J. Lin, Y.J. Lin, X.Z. Qi, Y.Y. Dai, L. Gu, J.S. Hu, S.F. Jin, Q. Shen, H. Wang, Nature 538 (2016) 84–87.
- [4] C.X. Zhu, G.M. Bollas, Appl. Catal. B: Environ. 235 (2018) 92–102.
- [5] X.-P. Fu, Q.-K. Shen, D. Shi, K. Wu, Z. Jin, X. Wang, R. Si, Q.-S. Song, C.-J. Jia, C.-H. Yan, Appl. Catal. B: Environ. 211 (2017) 176–187.
- [6] Y. Liu, J.F. Chen, Y. Zhang, RSC Adv. 5 (2015) 29002–29007.
- [7] J.-S. Bae, S.Y. Hong, J.C. Park, G.B. Rhim, M.H. Youn, H. Jeong, S.W. Kang, J.-I. Yang, H. Jung, D.H. Chun, Appl. Catal. B: Environ. 244 (2019) 576–582.
- [8] Y. Liu, J.F. Chen, J. Bao, Y. Zhang, ACS Catal. 5 (2015) 3905–3909.
- [9] T.A. Wezendonk, V.P. Santos, M.A. Nasalevich, Q.S.E. Warringa, I.A. Dugulan, A. Chojacki, A.C.J. Koeken, M. Ruitenbeek, G. Meima, H.U. Islam, G. Sankar, M. Makkee, F. Kapteijn, J. Gascon, ACS Catal. 6 (2016) 3236–3247.
- [10] H.M.T. Galvis, J.H. Bitter, C.B. Khare, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, Science 335 (2012) 835–838.
- [11] J. Wang, S.Y. Huang, S. Howard, B.W. Muir, H.Y. Wang, D.F. Kennedy, X.B. Ma, ACS Catal. 9 (2019) 7976–7983.
- [12] Y. Cheng, J. Lin, T.J. Wu, H. Wang, S.H. Xie, Y. Pei, S.R. Yan, M.H. Qiao, B.N. Zong, Appl. Catal. B: Environ. 204 (2017) 475–485.
- [13] H.Y. Suo, S.G. Wang, C.G. Zhang, J. Xu, B.S. Wu, Y. Yang, H.W. Xiang, Y.W. Li, J. Catal. 286 (2012) 111–123.
- [14] C. Yang, H.B. Zhao, Y.L. Hou, D. Ma, J. Am. Chem. Soc. 134 (2012) 15814–15821.
- [15] Y. Liu, F.X. Lu, Y. Tang, M.Y. Liu, F.F. Tao, Y. Zhang, Appl. Catal. B: Environ. 261 (2020) 118219.
- [16] Q. Chang, C.H. Zhang, C.W. Liu, Y.X. Wei, A.V. Cheruvathur, A.I. Dugulan, J.W. Niemantsverdriet, X.W. Liu, Y.R. He, M. Qing, L.R. Zheng, Y.F. Yun, Y. Yang, Y.W. Li, ACS Catal. 8 (2018) 3304–3316.
- [17] L.D. Mansker, Y.M. Jin, D.B. Bukur, A.K. Datye, Appl. Catal. A Gen. 186 (1999) 277–296.
- [18] P. Wang, W. Chen, F.K. Chiang, A.I. Dugulan, Y.J. Song, R. Pestman, K. Zhang, J.S. Yao, B. Feng, P. Miao, W. X, E.J.M. Hensen, Sci. Adv. 4 (2018) eaau2947.
- [19] J.H. Lee, H.-K. Lee, D.H. Chun, H. Choi, G.B. Rhim, M.H. Youn, H. Jeong, S.W. Kang, J. Yang, H. Jung, C.S. Kim, J.C. Park, Nano Res. 12 (2019) 2568–2575.
- [20] X.Z. Wang, C.H. Zhang, Q. Chang, L.C. Wang, B.L. Lv, J. Xu, H.W. Xiang, Y. Yang, Y.W. Li, Catal. Today 343 (2020) 91–100.
- [21] K. Cheng, V.V. Ordonsky, M. Virginie, B. Legras, P.A. Chernavskii, V.O. Kazak, C. Cordier, S. Paul, Y. Wang, A.Y. Khodakov, Appl. Catal. A Gen. 488 (2014) 66–77.
- [22] F. Jiang, M. Zhang, B. Liu, Y.B. Xu, X.H. Liu, Catal. Sci. Technol. 7 (2017) 1245–1265.
- [23] L.A. Cano, A.A.G. Blanco, G. Lener, S.G. Marchetti, K. Sapag, Catal. Today 282 (2017) 204–213.
- [24] H.G. Salazar-Contreras, A. Martínez-Hernández, A.A. Boix, G.A. Fuentes, E. Torres-García, Appl. Catal. B: Environ. 244 (2019) 414–426.
- [25] X.P. Zhou, J. Ji, D. Wang, X.Z. Duan, G. Qian, D. Chen, X.G. Zhou, Chem. Commun. 51 (2015) 8853–8856.
- [26] Q.J. Chen, G.G. Liu, S.Y. Ding, M.C. Sheikh, D.H. Long, Y. Yoneyama, N. Tsubaki, Chem. Eng. J. 334 (2018) 714–724.
- [27] E. de Smit, B.M. Weckhuysen, Chem. Soc. Rev. 37 (2008) 2758–2781.
- [28] S.Z. Li, W.P. Ding, G.D. Meitzner, E. Iglesia, J. Phys. Chem. B 106 (2002) 85–91.
- [29] E. de Smit, A.M. Beale, S. Nikitenko, B.M. Weckhuysen, J. Catal. 262 (2009) 244–256.
- [30] G.Z. Bian, A. Oonuki, N. Koizumi, H. Nomoto, M. Yamada, J. Mol. Catal. A Chem. 186 (2002) 203–213.
- [31] Z.P. Tian, C.G. Wang, Z. Si, L.L. Ma, L.G. Chen, Q.Y. Liu, Q. Zhang, H.Y. Huang, Appl. Catal. A Gen. 541 (2017) 50–59.
- [32] K.S.W. Sing, D.H. Everett, R.A.W. Haul, L. Moscou, R.A. Pierotti, J. Rouquerol, T. Siemieniowska, Pure Appl. Chem. 57 (1985) 603–619.
- [33] L.W. Niu, X.W. Liu, X.D. Wen, Y. Yang, J. Xu, Y.W. Li, Catal. Today 343 (2020) 101–111.
- [34] J. Zieliński, I. Zglinicka, L. Znak, Z. Kaszukur, Appl. Catal. A Gen. 381 (2010) 191–196.
- [35] M.Y. Ding, Y. Yang, B.S. Wu, Y.W. Li, T.J. Wang, L.L. Ma, Appl. Energy 160 (2015) 982–989.
- [36] K. Zhao, H. Tang, B.T. Qiao, L. Li, J. Wang, ACS Catal. 5 (2015) 3528–3539.
- [37] M.K. Gnanamani, G. Jacobs, H.H. Hamdeh, W.D. Shafer, F. Liu, S.D. Hopps, G.A. Thomas, B.H. Davis, ACS Catal. 6 (2016) 913–927.
- [38] S. Wang, T. Wu, J. Lin, Y. Ji, S. Yan, Y. Pei, S. Xie, B.N. Zong, M.H. Qiao, ACS Catal. 10 (2020) 6389–6401.
- [39] Y. Wei, L. Yan, C. Ma, C. Zhang, S. Sun, X. Wen, Y. Yang, Y.W. Li, ACS Appl. Nano Mater. 3 (2020) 7182–7191.
- [40] I. Vollmer, S. Ould-Chikh, A. Aguilar-Tapia, G. Li, E. Pidko, J.-L. Hazemann, F. Kapteijn, J. Gascon, J. Am. Chem. Soc. 141 (2019) 18814–18824.

- [41] X.F. Yu, Jian L. Zhang, X. Wang, Q.X. Ma, X.H. Gao, H.Q. Xia, X.Y. Lai, S.B. Fan, T.-S. Zhao, *Appl. Catal. B: Environ.* 232 (2018) 420–428.
- [42] B. Gu, V.V. Ordonsky, M. Bahri, O. Ersen, P.A. Chernavskii, D. Filimonov, A.Y. Khodakov, *Appl. Catal. B: Environ.* 234 (2018) 153–428.
- [43] Y.L. Zhang, D.L. Fu, X.L. Liu, Z.P. Zhang, C. Zhang, B.F. Shi, J. Xu, Y.F. Han, *ChemCatChem* 10 (2018) 1272–1276.
- [44] Z.P. Tian, C.G. Wang, J. Yue, X.H. Zhang, L.L. Ma, *Catal. Sci. Technol.* 9 (2019) 2728–2741.
- [45] F. Liu, J.H. Zhu, W.L. Yang, Y.H. Dong, Y.L. Hou, C.Z. Zhang, H. Yin, S.H. Sun, *Angew. Chem. Int. Ed.* 126 (2014) 2208–2212.
- [46] X.N. Li, K.Y. Zhu, J.F. Pang, M. Tian, J.Y. Liu, A.I. Rykov, M.Y. Zheng, X.D. Wang, X.F. Zhu, Y.Q. Huang, B. Liu, J.H. Wang, W.S. Yang, T. Zhang, *Appl. Catal. B: Environ.* 224 (2018) 518–532.
- [47] C. Yang, B. Zhao, R. Gao, S. Yao, P. Zhai, S. Li, J. Yu, Y.L. Hou, D. Ma, *ACS Catal.* 7 (2017) 5661–5667.
- [48] H.M.T. Galvis, J.H. Bitter, T. Davidian, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, *J. Am. Chem. Soc.* 134 (2012) 16207–16215.
- [49] C.F. Huo, Y.W. Li, J.G. Wang, H.J. Jiao, *J. Am. Chem. Soc.* 131 (2009) 14713–14721.
- [50] G.Z. Bian, A. Oonuki, N. Koizumi, H. Nomoto, M. Yamada, *J. Mol. Catal. A Chem.* 186 (2002) 203–213.
- [51] M. Xu, S. He, H. Chen, G.Q. Cui, L. Zheng, B. Wang, M. Wei, *ACS Catal.* 7 (2017) 7600–7609.
- [52] L.D. Deng, H. Miura, T. Shishido, Z. Wang, S. Hosokawa, K. Teramura, T. Tanaka, *J. Catal.* 365 (2018) 277–291.
- [53] L. DeRita, S. Dai, K. Lopez-Zepeda, N. Pham, G.W. Graham, X. Pan, P. Christopher, *J. Am. Chem. Soc.* 139 (2017) 14150–14165.
- [54] Y.M. Chen, B.C. Qiu, Y. Liu, Y. Zhang, *Appl. Catal. B: Environ.* 269 (2020) 118801.
- [55] X.L. Yan, B.R. Zhao, Y. Liu, Y. Li, *Catal. Today* 256 (2015) 29–40.
- [56] J. Saavedra, C.J. Pursell, B.D. Chandler, *J. Am. Chem. Soc.* 140 (2018) 3712–3723.
- [57] C.H. Mejia, J.E.Sv. Hoeven, P.E. de Jongh, K.P. de Jong, *ACS Catal.* 10 (2020) 7343–7354.
- [58] A.Y. Khodakov, W. Chu, P. Fongarland, *Chem. Rev.* 107 (2007) 1692–1744.
- [59] T.W.V. Deelen, C.H. Mejia, K.P. de Jong, *Nat. Catal.* 11 (2019) 955–970.
- [60] X. Fang, B. Liu, K. Cao, P. Yang, Q. Zhao, F. Jiang, Y. Xu, R. Chen, X.H. Liu, *ACS Catal.* 10 (2020) 2799–2816.

Revealing the activity of different iron carbides for Fischer-Tropsch synthesis

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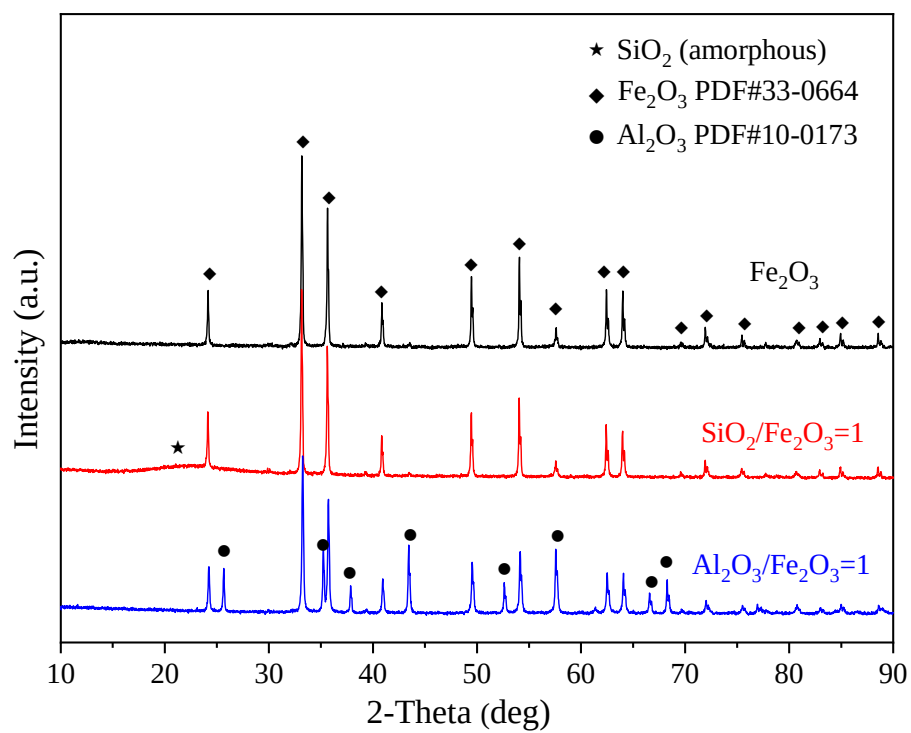


Figure S1. The XRD patterns of fresh catalysts.

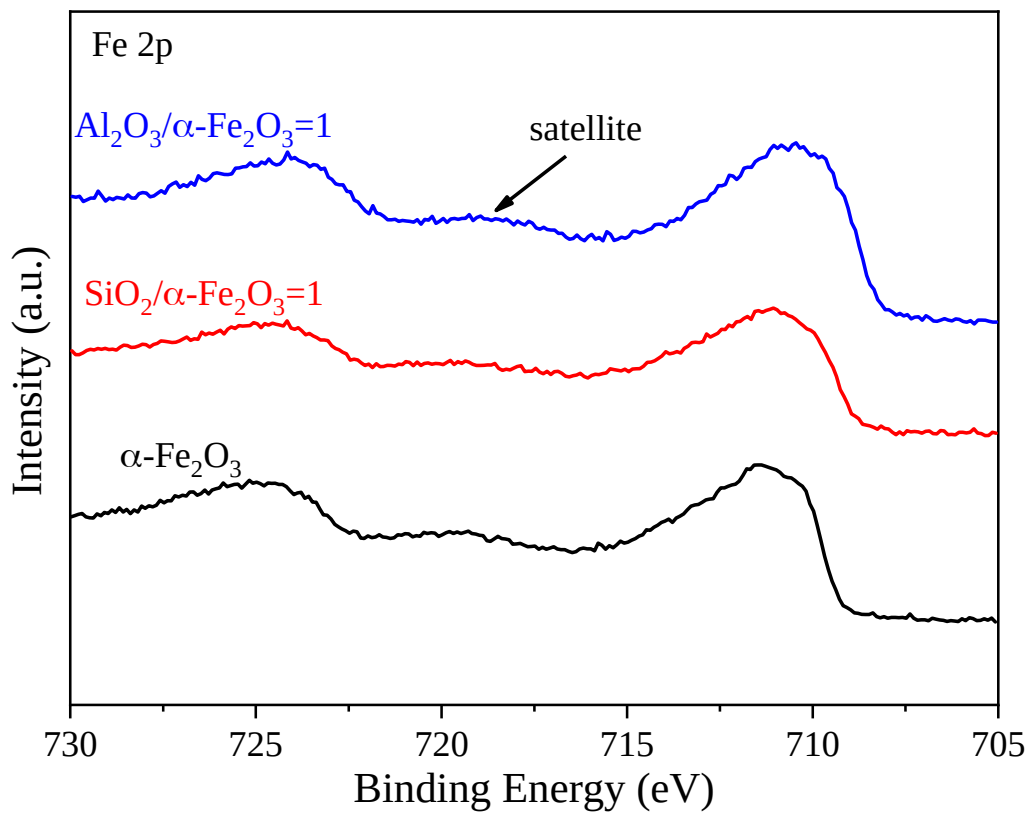


Figure S2. XPS spectra of Fe 2p on the fresh catalysts.

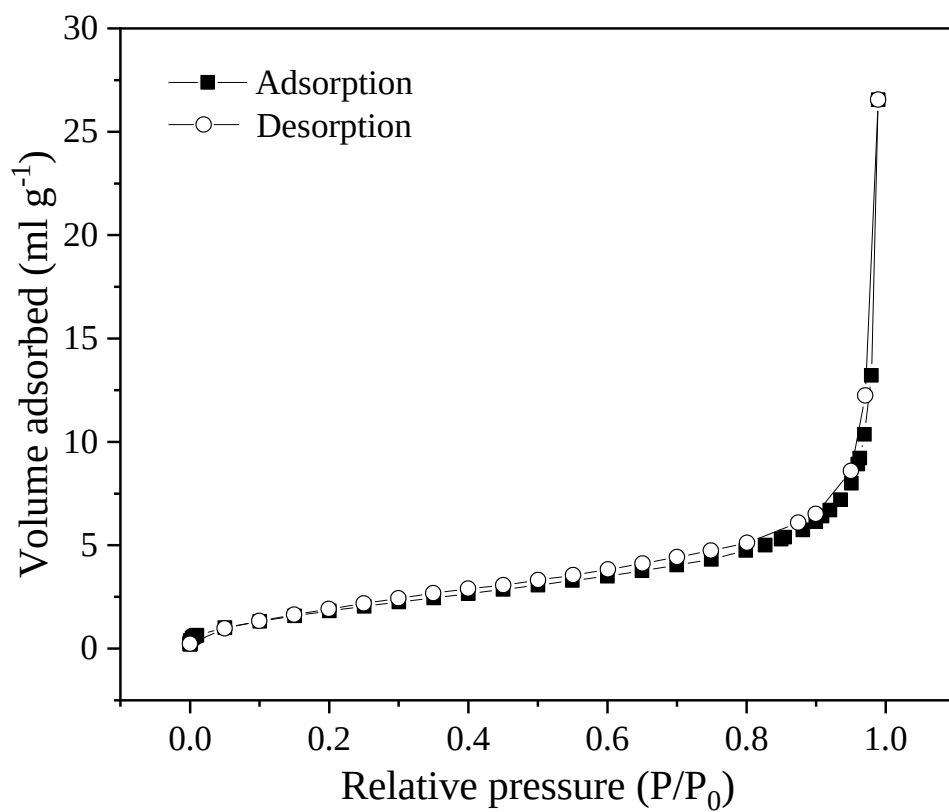


Figure S3. N₂ adsorption-desorption isotherms of the fresh α -Fe₂O₃ catalyst.

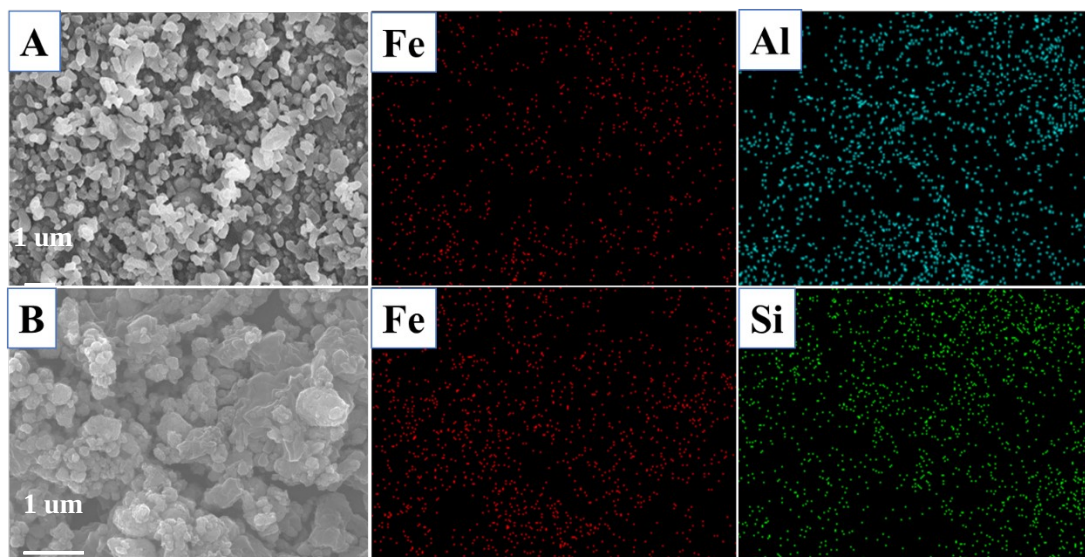


Figure S4 A) SEM-EDX elemental mapping of Fe and Al for the fresh Al₂O₃/α-Fe₂O₃=1 catalyst; B) SEM-EDX elemental mapping of Fe and Si for the fresh SiO₂/α-Fe₂O₃=1 catalyst.

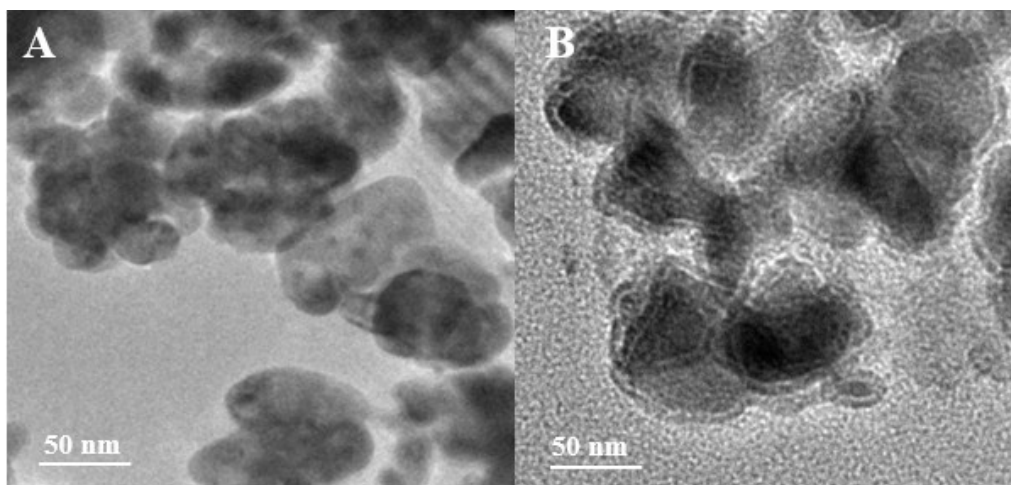


Figure S5 TEM images of the pure α -Fe₂O₃ catalyst before (A) and after (B) FTS reaction.

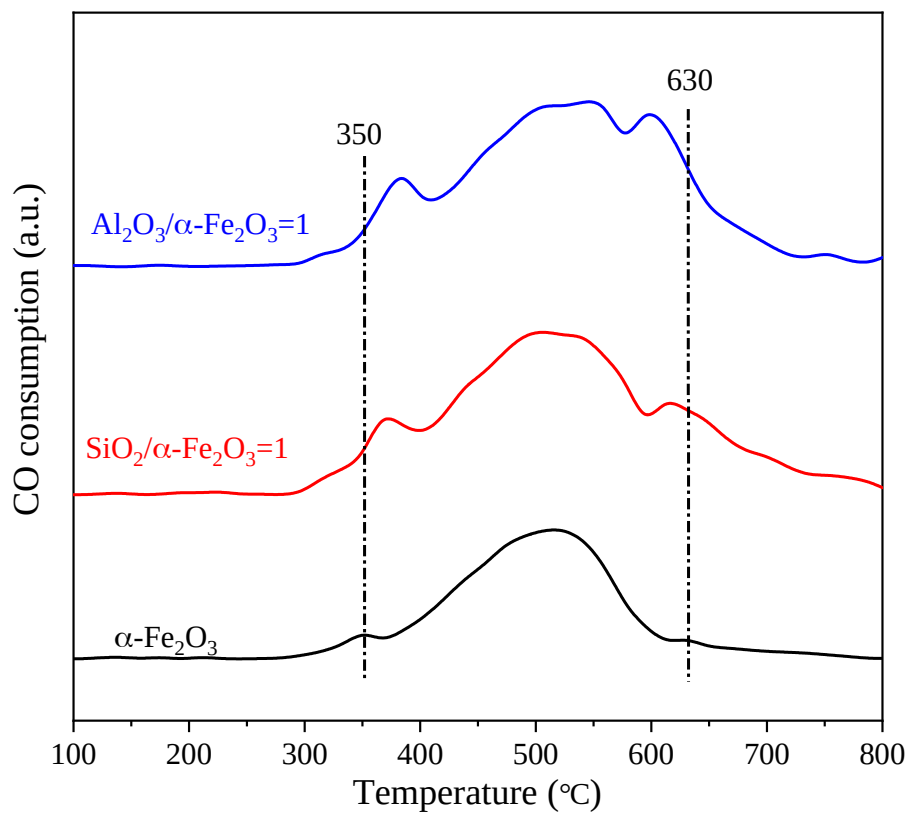


Figure S6 The CO-TPR profiles of various fresh catalysts for CO consumption.

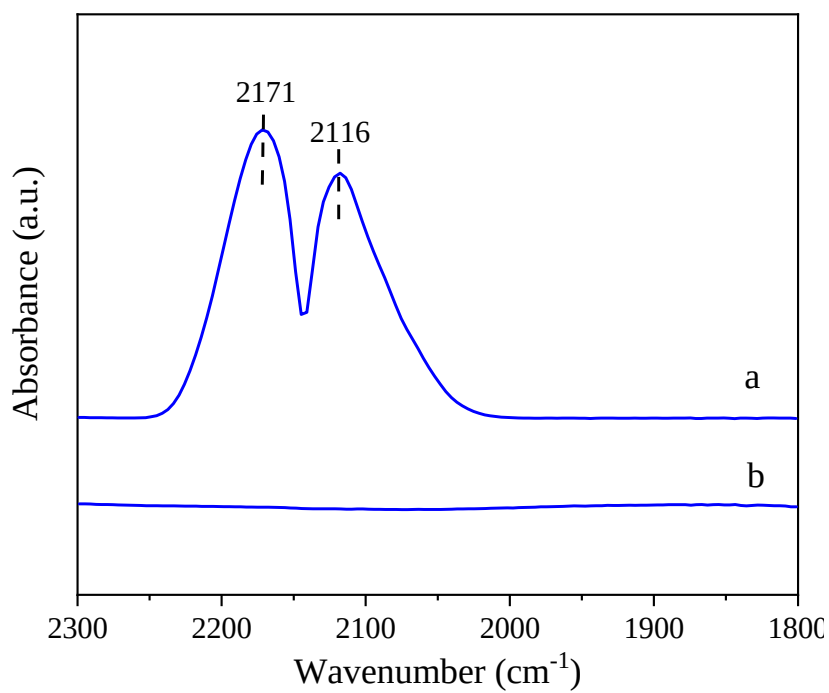
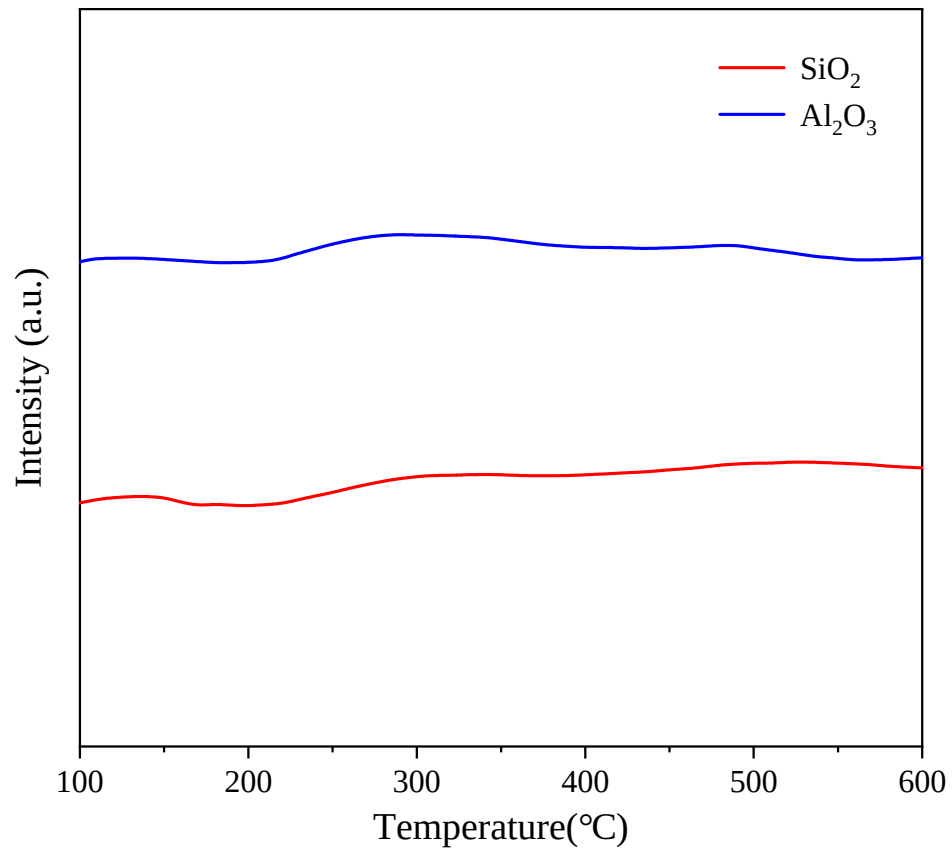


Figure S7 CO-DRIFT spectrum of the Al_2O_3 .

(a)



(b)

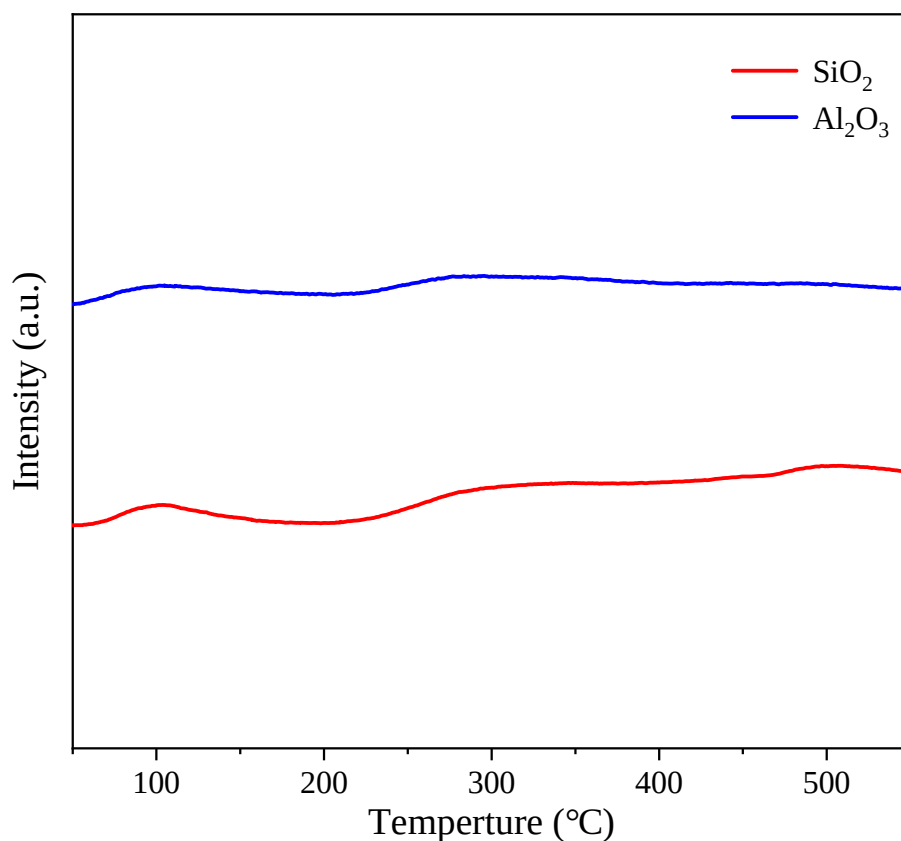


Figure S8 NH₃-TPD (a) and CO₂-TPD (b) of SiO₂ and Al₂O₃.

Table S1. The results of in-situ XPS for Fe 2p 3/2

catalysts	Binding energy (eV)				Iron carbide (atom%)			
	Fe 2p3/2				Fe 2p3/2			
	Fe ₂ C	Fe ₅ C ₂	Fe ₃ C	Fe ₂ O ₃	Fe ₂ C	Fe ₅ C ₂	Fe ₃ C	Fe ₂ O ₃
α-Fe ₂ O ₃	706.1	707.0	707.9	709.9	1.2	22.2	15.5	61.1
SiO ₂ /α-Fe ₂ O ₃ =1	706.7	707.3	708.3	710.4	7.1	19.3	11.7	61.9
Al ₂ O ₃ /α-Fe ₂ O ₃ =1	706.7	707.2	708.2	710.2	13.2	18.0	10.3	58.5

Table S2. The location and content of linear adsorbed CO on different iron carbides from in-situ CO-DRIFT.

catalysts	Wavenumber (cm ⁻¹)			Content of the adsorbed CO (%)		
	Fe ₂ C	Fe ₅ C ₂	Fe ₃ C	Fe ₂ C	Fe ₅ C ₂	Fe ₃ C
α-Fe ₂ O ₃	-	2013	2027	-	63.2	36.8

SiO ₂ /α-Fe ₂ O ₃ =1	2004	2014	2032	18.6	57.3	24.1
Al ₂ O ₃ /α-Fe ₂ O ₃ =1	2006	2016	2033	27.7	49.3	23.0
